Subject: Mathematics | **UNIT TEST – 01** | **Time:** 45 Minutes
Class: Class VIII (CBSE) | **Chapter 1: Fractions in Disguise** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 3 Short Answer Type questions of 2 marks each.
4. **Section C** comprises 3 Long Answer Type questions of 3 marks each.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. **The petrol price in 2015 was *Rs.*60 and *Rs.*100 in 2025. What is the percentage increase in the price of petrol?** [1]

(A) 40%
(B) 60%
(C) 66.66%
(D) 166.66%

Q2. **Jasmine invests an amount 'p' for 4 years at an interest rate of 6% p.a. Which of the following expressions describes the total amount she will get after 4 years when compounding is not done?** [1]

(A) $p \times 6 \times 4$
(B) $p \times 1.06 \times 4$
(C) $p + (p \times 0.06 \times 4)$
(D) $p \times \left(\frac{0.06}{100}\right) \times 4$

Q3. **If 30% of k is 70, then what is 90% of k?** [1]

(A) 140
(B) 210
(C) 270
(D) 280

Q4. **A number increased by 20% becomes 90. What is the original number?** [1]

(A) 70
(B) 72
(C) 75
(D) 108

Q5. **The population of Bengaluru in 2025 is about 250% of its population in 2000. If the population in 2000 was 50 lakhs, what is the population in 2025?** [1]

(A) 100 lakhs
(B) 125 lakhs
(C) 150 lakhs
(D) 75 lakhs

SECTION B: Short Answer Questions (3 × 2 = 6 Marks)

Q6. **A cyclist cycles from Delhi to Agra and completes 40% of the journey. If he has covered 92 km, how many more kilometres does he have to travel to reach Agra?** [2]

Q7. **A shopkeeper sells pencils at a price such that the selling price of 3 pencils is equal to the cost price of 5 pencils. Does he make a profit or a loss? Find his profit or loss percentage.** [2]

Q8. **In a laboratory, the number of bacteria in a certain experiment increases at the rate of 2.5% per hour. Find the number of bacteria at the end of 2 hours if the initial count is 5,06,000.** [2]

SECTION C: Long Answer Questions (3 × 3 = 9 Marks)

Q9. A milkman sold two buffaloes for *Rs.*80,000 each. On one of them, he made a profit of 5% and on the other a loss of 10%. Find his total cost price, total selling price, and overall profit or loss amount. (Round CP to the nearest rupee). [3]

Q10. A school is organising an excursion for its students. 40% of them are Grade 8 students and the rest are Grade 9 students. Among these Grade 8 students, 60% are girls. [3]

(i) What percentage of the total students going to the excursion are Grade 8 girls?

(ii) If the total number of students going to the excursion is 160, how many of them are Grade 8 girls, and how many are Grade 9 students?

Q11. Giridhar borrows a loan of *Rs.*12,500 at 12% per annum for 3 years without compounding, and Raghava borrows the same amount for the same time period at 10% per annum, compounded annually. Who pays more interest and by how much? [3]

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